

Alternating Current in Eight Derivations

Chapter 7 · Alternating Current — inside the 16-mark unit shared with EMI. Source: published page 1e4833b5-4f5c-490c-8b6f-b9bf9a3e972c. Plain-text maths, not KaTeX. Its eight derivations against the five (PD33–PD37) in Physics, Derived; this page also carries the theory, formula strip and question tiers.

Scope. No scope cut. Eight derivations, and this is the chapter most likely to supply a five-mark question on the paper.

01 · Theory

1 · Alternating current, mean value, rms value

An alternating current reverses direction periodically and varies in magnitude continuously, written $i = i_0 \sin \omega t$ with a matching voltage $v = v_0 \sin \omega t$. Here i_0 is the peak value or current amplitude, $\omega = 2\pi\nu$ the angular frequency.

Averaged over a **complete** cycle the current is zero — the negative half exactly cancels the positive half. That is why a moving-coil ammeter reads nothing on ac. Over **half** a cycle the average is not zero:

- Mean over half a cycle: $i_{\text{mean}} = 2i_0/\pi = 0.637 i_0$

Because the average is useless, ac is described by its **root mean square** value — the steady direct current that would produce the same heating in the same resistance over the same time. Since heating goes as i^2 , and the mean of $\sin^2 \omega t$ over a cycle is $\frac{1}{2}$:

- $I_{\text{rms}} = i_0/\sqrt{2} = 0.707 i_0$ and $V_{\text{rms}} = v_0/\sqrt{2} = 0.707 v_0$

Every ac meter reads rms, and every stated supply voltage — the 220 V mains — is an rms value. Its peak is $220\sqrt{2} \approx 311$ V.

2 · Reactance, impedance, power factor

- **Reactance** is the opposition offered by an inductor or a capacitor alone. Inductive reactance $X_L = \omega L$, capacitive reactance $X_C = 1/\omega C$. Both in ohm.
- **Impedance** Z is the total opposition of a circuit containing resistance and reactance together, $Z = V_{\text{rms}}/I_{\text{rms}}$, in ohm.
- **Phase angle** ϕ is the angle by which the current leads or lags the applied voltage.
- **Power factor** is $\cos \phi = R/Z$, a pure number between 0 and 1. It is the fraction of the apparent power $V_{\text{rms}} I_{\text{rms}}$ that is actually consumed.

Reactance differs from resistance in one crucial way: a resistor dissipates energy, a pure reactance does not. Over one cycle an inductor returns to the source exactly the energy it took, and so does a capacitor. That is why a purely reactive circuit consumes zero power however large the current.

3 • The three single elements side by side

Element	Opposition	Phase of current	Against frequency	Power
Resistor R	R, constant	In phase with V	Independent of ν	$V_{\text{rms}} I_{\text{rms}}$
Inductor L	$X_L = \omega L$	Lags V by $\pi/2$	Straight line through origin	Zero
Capacitor C	$X_C = 1/\omega C$	Leads V by $\pi/2$	Rectangular hyperbola	Zero

A memory hook the examiner expects you to use correctly: **CIVIL** — in a **C**apacitor **I** comes before **V**; **V** comes before **I** in an inductor **L**.

Two consequences worth knowing as one-liners: a capacitor blocks dc (at $\nu = 0$, $X_C = \infty$) but passes ac; an inductor passes dc freely (at $\nu = 0$, $X_L = 0$) but chokes ac, which is what a choke coil is for.

4 • Resonance and sharpness

In a series LCR circuit X_L rises with frequency and X_C falls, so at one frequency they are equal and cancel. There the impedance collapses to its smallest possible value, $Z = R$, the current reaches its largest possible value v_0/R , and voltage and current come into phase. That is **resonance**, at $\omega_0 = 1/\sqrt{LC}$.

Resonance only exists if both L and C are present — with one of them missing there is nothing to cancel.

Sharpness is how quickly the current falls away on either side of ω_0 . It is measured by the quality factor

- $Q = \omega_0 L/R = (1/R)\sqrt{L/C} = \omega_0/(\omega_2 - \omega_1)$, where ω_1 and ω_2 are the half-power frequencies at which the current has dropped to $1/\sqrt{2}$ of its peak.

Smaller R means larger Q, a taller and narrower peak, and a circuit that responds to a narrower band of frequencies — which is exactly what tuning a radio to one station requires.

5 • Transformer, and why transmission uses high voltage

A transformer changes an alternating voltage up or down. It works on **mutual induction**: alternating current in the primary sets up a changing flux in a shared soft-iron core, and that changing flux induces an alternating emf of the same frequency in the secondary. It cannot work on dc, because a steady current makes no changing flux.

The transmission argument, which is asked directly:

- A power station must deliver a fixed power $P = VI$ down a line of resistance R.
- The heat wasted in the line is $I^2 R$.
- For a fixed P, stepping the voltage up divides the current by the same factor.
- Since the loss goes as I^2 , ten times the voltage means a hundredth of the loss.
- So a step-up transformer at the generating end raises the voltage for the journey, and step-down transformers at the consumer end bring it back to 220 V.

A step-up transformer does not create energy. Raising the voltage lowers the current by the same factor, so $V_P I_P = V_S I_S$ still holds.

02 · Derivations

D1 · AC through a pure resistor

A source of alternating voltage $v = v_0 \sin \omega t$ is connected across a pure resistance R , with no inductance or capacitance in the circuit.

1. $v = v_0 \sin \omega t$
2. Kirchhoff's loop rule: $v - iR = 0$
3. $i = v / R$
4. $i = (v_0 \sin \omega t) / R$
5. $i = (v_0/R) \sin \omega t$
6. Compare with $i = i_0 \sin(\omega t + \phi)$
7. $i_0 = v_0 / R$ and $\phi = 0$
8. $\phi = 0 \Rightarrow$ current and voltage are in phase
9. $i^2 = i_0^2 \sin^2 \omega t$
10. Mean of $\sin^2 \omega t$ over one full cycle = $1/2$
11. Mean of $i^2 = i_0^2 / 2$
12. $I_{\text{rms}} = \sqrt{(i_0^2/2)} = i_0/\sqrt{2} = 0.707 i_0$
13. Likewise $V_{\text{rms}} = v_0/\sqrt{2}$
14. Dividing: $V_{\text{rms}} / I_{\text{rms}} = (v_0/\sqrt{2}) / (i_0/\sqrt{2}) = v_0/i_0 = R$
15. Instantaneous power $p = vi = (v_0 \sin \omega t)(i_0 \sin \omega t)$
16. $p = v_0 i_0 \sin^2 \omega t$
17. Average over a cycle: $P = v_0 i_0 \times (1/2)$
18. $P = (v_0/\sqrt{2})(i_0/\sqrt{2})$
19. $P = V_{\text{rms}} I_{\text{rms}}$

Result: $i = (v_0/R) \sin \omega t$, $\phi = 0$ · $I_{\text{rms}} = i_0/\sqrt{2}$ · $P = V_{\text{rms}} I_{\text{rms}}$

Graph and phasor: v and i are sine curves that cross zero together and peak together. On the phasor diagram V_0 and i_0 lie along the same line, angle between them zero.

D2 · AC through a pure inductor — $X_L = \omega L$

The same source $v = v_0 \sin \omega t$ across a coil of pure self-inductance L with negligible resistance.

1. $v = v_0 \sin \omega t$
2. Induced back emf in the coil: $\varepsilon = -L di/dt$
3. Kirchhoff's loop rule: $v + \varepsilon = 0$
4. $v - L (di/dt) = 0$
5. $L (di/dt) = v_0 \sin \omega t$
6. $di/dt = (v_0/L) \sin \omega t$
7. $di = (v_0/L) \sin \omega t dt$
8. $\int di = (v_0/L) \int \sin \omega t dt$

9. $i = (v_0/L) (-\cos \omega t / \omega) + \text{constant}$
10. The source has no steady component, so the constant of integration = 0
11. $i = -(v_0/\omega L) \cos \omega t$
12. $-\cos \omega t = \sin(\omega t - \pi/2)$
13. $i = (v_0/\omega L) \sin(\omega t - \pi/2)$
14. Compare with $i = i_0 \sin(\omega t - \pi/2)$: $i_0 = v_0 / (\omega L)$
15. Compare with $i_0 = v_0 / X_L$
16. $X_L = \omega L = 2\pi\nu L$
17. Phase of current = $\omega t - \pi/2$; phase of voltage = ωt
18. Current phase is behind voltage phase by $\pi/2$

Result: $X_L = \omega L = 2\pi\nu L \cdot i = i_0 \sin(\omega t - \pi/2) \cdot \text{current lags } V \text{ by } \pi/2$

Graph and phasor: X_L against ν is a straight line through the origin. On the phasor diagram V_0 points up and i_0 is 90° clockwise behind it. On the v - i time graph the current curve peaks a quarter cycle after the voltage curve.

D3 · AC through a pure capacitor — $X_C = 1/\omega C$

The same source $v = v_0 \sin \omega t$ across a capacitor of capacitance C .

Shared setup: this is derivation 2 with one substitution. Step 1 is identical, steps 12-18 of D2 map one-to-one onto steps 8-13 here, and the boxed result has the same shape. The only real difference: the inductor needs an *integration* and the capacitor a *differentiation*, and that flips the sign of the $\pi/2$.

1. $v = v_0 \sin \omega t$
2. Charge on the capacitor at any instant: $q = Cv$
3. $q = C v_0 \sin \omega t$
4. $i = dq/dt$
5. $i = d/dt (C v_0 \sin \omega t)$
6. $i = C v_0 \omega \cos \omega t$
7. $i = [v_0 / (1/\omega C)] \cos \omega t$
8. $\cos \omega t = \sin(\omega t + \pi/2)$
9. $i = [v_0 / (1/\omega C)] \sin(\omega t + \pi/2)$
10. Compare with $i = i_0 \sin(\omega t + \pi/2)$: $i_0 = v_0 / (1/\omega C)$
11. Compare with $i_0 = v_0 / X_C$
12. $X_C = 1/\omega C = 1/(2\pi\nu C)$
13. Current phase ($\omega t + \pi/2$) is ahead of voltage phase (ωt) by $\pi/2$

Result: $X_C = 1/\omega C = 1/(2\pi\nu C) \cdot i = i_0 \sin(\omega t + \pi/2) \cdot \text{current leads } V \text{ by } \pi/2$

Graph and phasor: X_C against ν is a rectangular hyperbola, falling steeply and never touching either axis. On the phasor diagram i_0 is 90° ahead of V_0 . At $\nu = 0$ the reactance is infinite, which is the statement that a capacitor blocks dc.

D4 • Series LCR circuit by phasor diagram

R, L and C in series across $v = v_0 \sin \omega t$. Being in series, the same instantaneous current flows through all three, so the current is taken as the reference phasor.

1. Same current I through R, L and C
2. Draw the current phasor I along the reference axis
3. $V_R = I R$, in phase with I — drawn along I
4. $V_L = I X_L$, leading I by $\pi/2$ — drawn 90° ahead
5. $V_C = I X_C$, lagging I by $\pi/2$ — drawn 90° behind
6. V_L and V_C are opposite in direction, so they partly cancel
7. Their resultant has magnitude $(V_L - V_C)$, along V_L when $V_L > V_C$
8. V_R and $(V_L - V_C)$ are perpendicular
9. The applied voltage V is the resultant of these two perpendicular phasors
10. $V^2 = V_R^2 + (V_L - V_C)^2$
11. $V = \sqrt{V_R^2 + (V_L - V_C)^2}$
12. Substituting $V_R = IR$, $V_L = IX_L$, $V_C = IX_C$:
13. $V = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$
14. $V = \sqrt{I^2[R^2 + (X_L - X_C)^2]}$
15. $V = I \sqrt{R^2 + (X_L - X_C)^2}$
16. Impedance is defined as $Z = V / I$
17. $Z = \sqrt{R^2 + (X_L - X_C)^2}$
18. Substituting $X_L = \omega L$ and $X_C = 1/\omega C$:
19. $Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$
20. From the phasor triangle, $\tan \phi = (V_L - V_C) / V_R$
21. $\tan \phi = (I X_L - I X_C) / (I R)$
22. $\tan \phi = (X_L - X_C) / R$
23. $\phi = \tan^{-1}[(X_L - X_C) / R]$
24. Peak current $i_0 = v_0 / Z$
25. $i = [v_0 / \sqrt{R^2 + (\omega L - 1/\omega C)^2}] \sin(\omega t - \phi)$

Result: $Z = \sqrt{R^2 + (X_L - X_C)^2}$ · $\tan \phi = (X_L - X_C)/R$

Phasor diagram: current along the horizontal reference axis; V_R along it; V_L straight up; V_C straight down; $(V_L - V_C)$ as the surviving vertical; the hypotenuse from the origin is V , and the angle it makes with V_R is ϕ .

$X_L > X_C$: ϕ positive, circuit is inductive, current lags. $X_C > X_L$: ϕ negative, circuit is capacitive, current leads. $X_L = X_C$: $\phi = 0$, resonance.

D5 • Resonance in a series LCR circuit

The circuit of derivation 4, with R, L and C fixed and the frequency ω of the source varied. Starts from the boxed result of D4, so quote that line first in the exam.

1. From D4: $Z = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$

2. $i_0 = v_0 / Z$
3. v_0 is fixed, so i_0 is maximum when Z is minimum
4. R is fixed; the only term that changes with ω is $(\omega L - 1/\omega C)^2$
5. $(\omega L - 1/\omega C)^2 \geq 0$ for every ω , and its least value is 0
6. Z is minimum when $\omega L - 1/\omega C = 0$
7. $\omega L = 1/\omega C$
8. $\omega^2 LC = 1$
9. $\omega^2 = 1/(LC)$
10. $\omega_0 = 1/\sqrt{LC}$
11. $v_0 = \omega_0/2\pi = 1/(2\pi\sqrt{LC})$
12. At $\omega = \omega_0$: $X_L = X_C$, so $X_L - X_C = 0$
13. $Z_{\min} = \sqrt{R^2 + 0} = R$
14. $i_0(\max) = v_0 / R$
15. From D4: $\tan \phi = (X_L - X_C)/R = 0/R = 0$
16. $\phi = 0 \Rightarrow$ current and applied voltage are in phase
17. The circuit behaves as though only R were present

Result: $\omega_0 = 1/\sqrt{LC}$, $v_0 = 1/(2\pi\sqrt{LC}) \cdot Z = R \cdot i_0(\max) = v_0/R$

Graph: i_0 against ν rises to a peak at ν_0 and falls away on both sides. Draw two curves for two values of R : the smaller R gives the taller, sharper peak. Mark the half-power frequencies ν_1 and ν_2 where the current is $i_0(\max)/\sqrt{2}$, and label the width $(\nu_2 - \nu_1)$.

Sharpness is measured by $Q = \omega_0 L/R = (1/R)\sqrt{L/C} = \omega_0/(\omega_2 - \omega_1)$. Resonance is impossible unless both L and C are present.

D6 · Average power in an AC circuit, and wattless current

A voltage $v = v_0 \sin \omega t$ drives a current $i = i_0 \sin(\omega t - \phi)$ through any ac circuit, ϕ being the phase difference between them.

1. Instantaneous power $p = v i$
2. $p = v_0 \sin \omega t \cdot i_0 \sin(\omega t - \phi)$
3. $p = v_0 i_0 \sin \omega t \sin(\omega t - \phi)$
4. $p = (v_0 i_0 / 2) \cdot 2 \sin \omega t \sin(\omega t - \phi)$
5. Identity: $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
6. Here $A = \omega t$ and $B = \omega t - \phi$
7. $A - B = \phi$ and $A + B = 2\omega t - \phi$
8. $2 \sin \omega t \sin(\omega t - \phi) = \cos \phi - \cos(2\omega t - \phi)$
9. $p = (v_0 i_0 / 2)[\cos \phi - \cos(2\omega t - \phi)]$
10. Average over one complete cycle: $P = (v_0 i_0 / 2)[\cos \phi - \langle \cos(2\omega t - \phi) \rangle]$
11. $\langle \cos(2\omega t - \phi) \rangle$ over one complete cycle = 0
12. $P = (v_0 i_0 / 2) \cos \phi$
13. $P = (v_0/\sqrt{2})(i_0/\sqrt{2}) \cos \phi$
14. $P = V_{\text{rms}} I_{\text{rms}} \cos \phi$

15. Purely resistive: $\phi = 0$, $\cos \phi = 1$, $P = V_{\text{rms}} I_{\text{rms}}$ — maximum
16. Purely inductive or purely capacitive: $\phi = \pi/2$, $\cos \phi = 0$
17. $P = V_{\text{rms}} I_{\text{rms}} \times 0 = 0$
18. Current flows but no power is consumed — the current is wattless
19. Wattless component of the current $= I_{\text{rms}} \sin \phi$

Result: $P = V_{\text{rms}} I_{\text{rms}} \cos \phi$ · $\cos \phi = R/Z$ is the power factor

Graph: plot v , i and p on one set of axes. The power curve oscillates at twice the frequency of v and i ; when $\phi = \pi/2$ it spends equal time above and below the axis, so its average is zero.

Note that $P = 0$ for a pure reactance is not because the resistance is zero, but because voltage and current are 90° out of phase.

D7 · Transformer — turns ratio, and the four losses

Two coils, N_P turns in the primary and N_S in the secondary, wound on a common laminated soft-iron core. Assume no flux leakage, so the same flux ϕ links every turn of both coils, and negligible primary resistance.

1. Flux through each turn of either coil $= \phi$
2. Faraday's law for the primary: $\varepsilon_P = -N_P (d\phi/dt)$
3. Faraday's law for the secondary: $\varepsilon_S = -N_S (d\phi/dt)$
4. Divide the second by the first: $\varepsilon_S / \varepsilon_P = [-N_S (d\phi/dt)] / [-N_P (d\phi/dt)]$
5. $\varepsilon_S / \varepsilon_P = N_S / N_P$
6. Primary resistance negligible $\Rightarrow \varepsilon_P = V_P$
7. Secondary on open circuit $\Rightarrow \varepsilon_S = V_S$
8. $V_S / V_P = N_S / N_P = r$ — ($r = \text{turns ratio}$)
9. For an ideal transformer, output power = input power
10. $V_S I_S = V_P I_P$
11. $I_P / I_S = V_S / V_P$
12. $I_P / I_S = N_S / N_P = r$
13. $I_S / I_P = N_P / N_S = 1/r$

Result: $V_S/V_P = N_S/N_P = I_P/I_S$

$N_S > N_P$ is a step-up transformer: voltage rises, current falls. $N_S < N_P$ is a step-down: voltage falls, current rises. Efficiency $= (V_S I_S / V_P I_P) \times 100\%$.

The four energy losses, and the fix for each

Loss	What is happening	The fix
Copper loss	Joule heating i^2R in the primary and secondary windings	Use thick copper wire of low resistance
Iron loss (eddy currents)	The changing flux induces circulating currents in the solid iron core, which heat it	Laminate the core — thin sheets, insulated from one another, so the eddy loops are broken
Flux leakage	Not all the flux made by the primary reaches the secondary	Wind the two coils one over the other on the same core
Hysteresis loss	The core is magnetised and demagnetised every cycle and energy is dissipated each time round the loop	Use a soft magnetic material — soft iron or silicon steel — with a thin hysteresis loop

Xam Idea adds a fifth, humming loss — energy lost as sound when the core vibrates. Give the four above unless the question asks for five.

D8 · AC generator — the emf equation

A coil of N turns, each of area A , rotates with constant angular speed ω about an axis perpendicular to a uniform magnetic field B . At $t = 0$ the plane of the coil is perpendicular to the field, so the area vector is along B .

1. Angle between B and the area vector at time t : $\theta = \omega t$
2. Flux through one turn: $\phi = B A \cos \theta$
3. $\phi = B A \cos \omega t$
4. Flux linkage through N turns: $N\phi = N B A \cos \omega t$
5. Faraday's law: $\varepsilon = -d(N\phi)/dt$
6. $\varepsilon = -d/dt (N B A \cos \omega t)$
7. N , B and A are constants: $\varepsilon = -N B A \cdot d/dt (\cos \omega t)$
8. $d/dt (\cos \omega t) = -\omega \sin \omega t$
9. $\varepsilon = -N B A (-\omega \sin \omega t)$
10. $\varepsilon = N B A \omega \sin \omega t$
11. $\sin \omega t$ has maximum magnitude 1
12. Peak emf $\varepsilon_0 = N B A \omega$
13. $\varepsilon = \varepsilon_0 \sin \omega t$
14. With $\omega = 2\pi\nu$: $\varepsilon = \varepsilon_0 \sin 2\pi\nu t$

Result: $\varepsilon = NBA\omega \sin \omega t = \varepsilon_0 \sin \omega t$, with $\varepsilon_0 = NBA\omega$

Graph: ε against t is a sine curve, one complete cycle per revolution of the coil, zero when the coil plane is perpendicular to B and peak when the plane is parallel to B .

Four parts to name in the diagram: field magnet, armature coil, slip rings, carbon brushes. The output can be raised by increasing N , increasing B , increasing A , or spinning faster. A moving-coil galvanometer cannot read this current, because its average over a cycle is zero.

03 · Formula strip

● *must be instant* · ○ *a few seconds is fine*

● rms value of an alternating current or voltage

$I_{\text{rms}} = i_0/\sqrt{2} = 0.707 i_0$ · $V_{\text{rms}} = v_0/\sqrt{2}$ — i_0, v_0 = peak values. I in ampere (A), V in volt (V). This is what every ac meter reads.

○ Mean value of an alternating current over half a cycle

$i_{\text{mean}} = 2i_0/\pi = 0.637 i_0$ — in ampere (A). Over a *full* cycle the mean is zero.

● Inductive reactance

$X_L = \omega L = 2\pi\nu L$ — L = self-inductance in henry (H), ν = frequency in hertz (Hz), $\omega = 2\pi\nu$ in rad s^{-1} . X_L in ohm (Ω).

● Capacitive reactance

$X_C = 1/\omega C = 1/(2\pi\nu C)$ — C = capacitance in farad (F). X_C in ohm (Ω). Infinite at $\nu = 0$, which is why a capacitor blocks dc.

● Impedance of a series LCR circuit

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ — R, X_L, X_C, Z all in ohm (Ω). Reduces to R alone at resonance.

● Phase angle of a series LCR circuit

$\tan \phi = (X_L - X_C)/R$ — ϕ in radian or degree. Positive ϕ means the current lags; negative means it leads.

● Power factor, in terms of the circuit constants

$\cos \phi = R / Z$ — a pure number between 0 and 1. Equal to 1 for a pure resistance and at resonance, 0 for a pure reactance.

● Average power consumed in any ac circuit

$P = V_{\text{rms}} I_{\text{rms}} \cos \phi$ — P in watt (W). $V_{\text{rms}} I_{\text{rms}}$ alone is the apparent power; only the $\cos \phi$ fraction of it is consumed.

○ Wattless component of the current

$I_{\text{wattless}} = I_{\text{rms}} \sin \phi$ — in ampere (A). This component transfers no net energy over a cycle.

● Resonant frequency of a series LCR circuit

$\omega_0 = 1/\sqrt{LC}$ · $\nu_0 = 1/(2\pi\sqrt{LC})$ — L in henry (H), C in farad (F). ω_0 in rad s^{-1} , ν_0 in hertz (Hz).

○ **Quality factor, all three forms**

$Q = \omega_0 L/R = (1/R)\sqrt{L/C} = \omega_0/(\omega_2 - \omega_1)$ — a pure number, no unit. ω_1 and ω_2 are the half-power frequencies. Larger Q means a sharper resonance.

● **Transformer equation, all three ratios**

$V_S/V_P = N_S/N_P = I_P/I_S$ — N = number of turns (no unit), V in volt (V), I in ampere (A). Note the current ratio is inverted.

○ **Efficiency of a transformer**

$\eta = (V_S I_S) / (V_P I_P) \times 100\%$ — output power over input power, as a percentage. Always below 100% in practice.

● **emf generated by an ac generator**

$\varepsilon = NBA\omega \sin \omega t = \varepsilon_0 \sin \omega t$, $\varepsilon_0 = NBA\omega$ — N = turns, B in tesla (T), A in m^2 , ω in $rad\ s^{-1}$. ε in volt (V).

○ **Net voltage across a series LCR circuit from the three drops**

$V = \sqrt{V_R^2 + (V_L - V_C)^2}$ — all in volt (V). This is why the three readings can add to more than the source voltage — they add as phasors, not as numbers.

○ **Power wasted as heat in a transmission line**

$P_{loss} = I^2 R$ — I = line current in ampere (A), R = total line resistance in ohm (Ω), P_{loss} in watt (W). Ten times the voltage means a hundredth of the loss.

04 · Questions to attempt

Question numbers only, from Xam Idea Chapter 7 and the NCERT questions reprinted inside it. Section names are Xam Idea's own.

Tier 1 — must do

15 questions · these are the 3-mark and 5-mark slots

Question	Page	Why
Long Answer Q1	p. 249	Inductor derivation, full
Long Answer Q2	p. 250	LCR phasor, both variants
Long Answer Q3	p. 252	Capacitor derivation, disguised
Long Answer Q4	p. 253	Average power derivation
Long Answer Q5	p. 253	Resonance plus graph
Long Answer Q6	p. 254	Generator, three variants
Long Answer Q7	p. 255	Transformer plus losses
Long Answer Q9	p. 258	Sharpness and Q
Short Answer Q1	p. 241	Capacitor in 3 marks
Short Answer Q2	p. 241	Minimum Z, wattless
Short Answer Q3	p. 241	Transformer plus transmission
Short Answer Q6	p. 242	The voltage paradox
Short Answer Q8	p. 243	Resonance numerical, complete
Short Answer Q18	p. 247	Three drops, phasor sum
NCERT Q8	p. 230	Everything at once

Tier 2 — if time

extra pattern coverage, mostly 1 and 2 marks

Question	Page	Why
NCERT Q1-Q7	p. 228-229	rms and power drills
Assertion-Reason 1-10	p. 234	Whole set, ten minutes
MCQ 7, 8, 16	p. 231-232	Average-power one-markers
MCQ 11, 12, 13, 17, 25, 26	p. 231-233	LCR and phasor reasoning
MCQ 3, 15, 18, 22, 23, 24	p. 231-233	Waveform and reactance graphs
MCQ 5, 10, 21	p. 231-232	Transformer and transmission
Very Short Ans Q1, Q3-Q6	p. 236-237	Definitions, written out
Very Short Ans Q11	p. 238	Black box plus phasor
Very Short Ans Q12, Q13	p. 238	Generator numericals
Very Short Ans Q19, Q20	p. 240	Transformer, line loss
Short Answer Q4, Q7, Q12	p. 242-244	Brightness reasoning
Short Answer Q5, Q9, Q10, Q13	p. 242-245	Impedance numericals
Short Answer Q11, Q19	p. 244, 248	Transformer numericals
Short Answer Q14, Q16, Q17	p. 246-247	Resonance and graphs
Long Answer Q8	p. 257	Reasoning plus numerical

Tier 3 — skip unless revising

off-pattern, over-difficult, or a repeat of something above

Off the pattern of this unit: Case Study Q1 (p. 235) is an RC charging transient — exponential decay in a dc circuit, not an alternating-current question. Very Short Answer Q9 and Q10 (p. 237) derive mean and rms values by integration; the paper will give you the results, not ask you to integrate them.

Duplicates — do one of each pair, not both: NCERT Q3 $\equiv$ Very Short Answer Q17. NCERT Q7 $\equiv$ Short Answer Q8. NCERT Q8 $\equiv$ Short Answer Q14. NCERT Q6 is just the resonance formula with numbers in it.

Already covered by the theory section: Very Short Answer Q2, Q7, Q8, Q15, Q16 and Short Answer Q15 are pure recall you will have from Section 01. Read the answers, do not write them out.

The five numerical types this chapter can ask

1. **Peak to rms, and the power a single element takes** — $I_{\text{rms}} = i_0/\sqrt{2}$, then $P = V_{\text{rms}} I_{\text{rms}} \cos \phi$
2. **Reactance of a coil or capacitor at a stated frequency** — $X_L = 2\pi\nu L$ · $X_C = 1/(2\pi\nu C)$
3. **Impedance, current and phase angle of a series LCR circuit** — $Z = \sqrt{R^2 + (X_L - X_C)^2}$
4. **Resonance: the frequency, the peak current, the individual voltage drops** — $\omega_0 = 1/\sqrt{LC}$, with $Z = R$ there
5. **Transformer turns and currents, or loss in a transmission line** — $V_S/V_P = N_S/N_P$
 $= I_P/I_S$, and $P_{\text{loss}} = I^2 R$